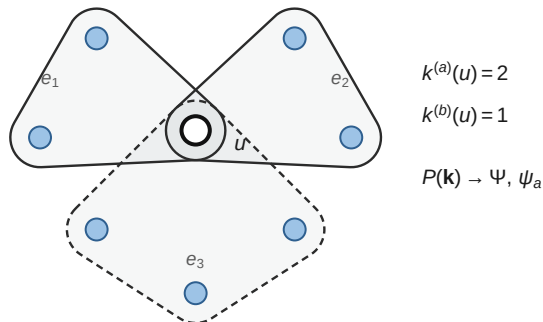
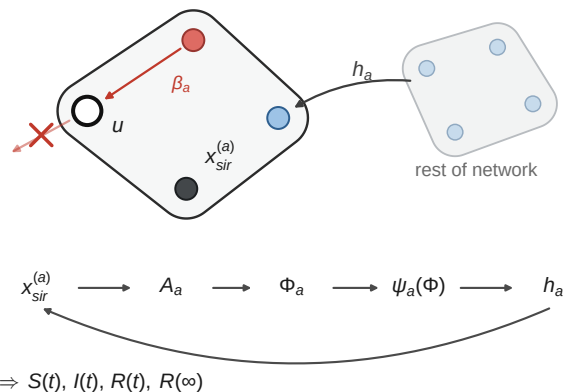


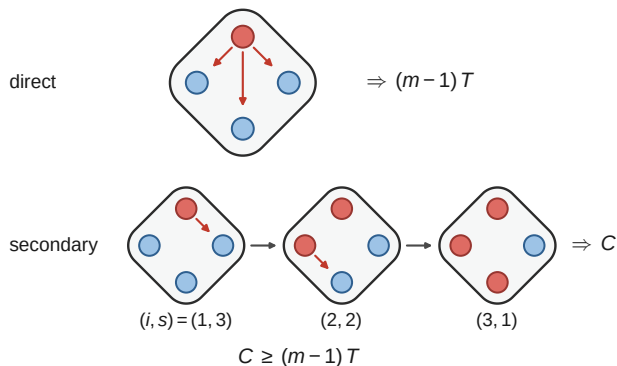
(a) Multiplex hypergraph



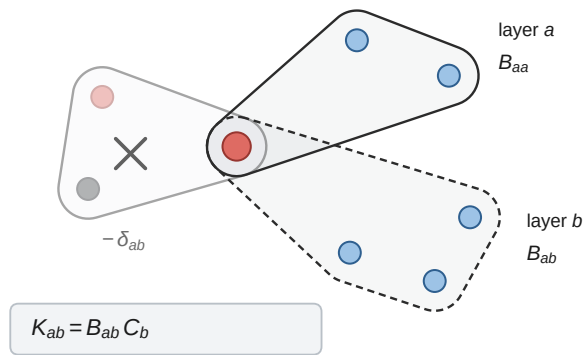
(b) Edge-based closure



(c) Within-hyperedge factor C



(d) Excess hyperdegree B



● susceptible
 ● infected
 ● recovered
 ○ test node u

layer a
 layer b